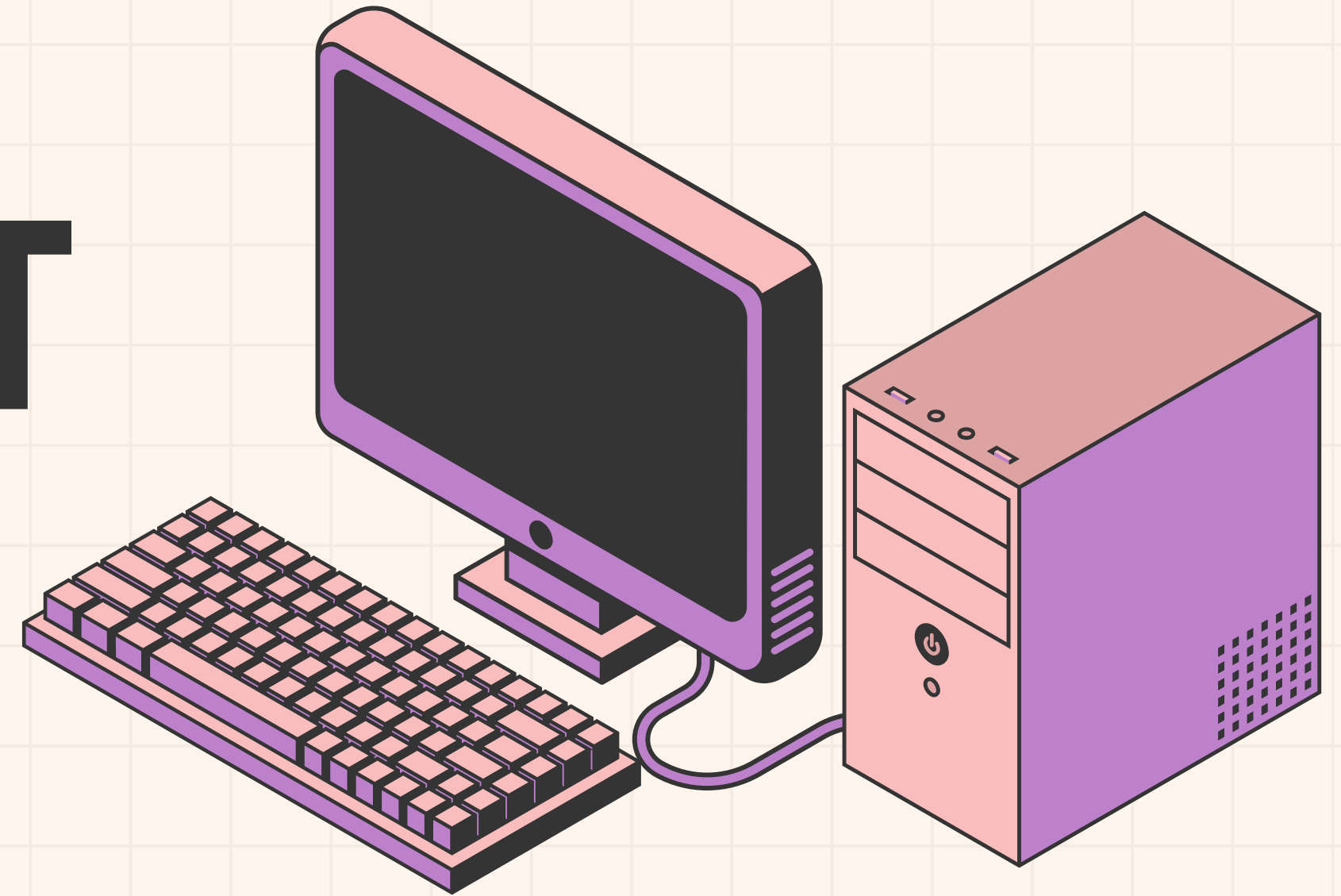


TASK MANAGMENT SYSTEM



SE-2527: Aziza Kenzhegali, Aldiyar Temkeshev, Daniyar Baikanov, Mansur Nursultanov

HOW DO WE SEE THE SOLUTION?

We see the solution as a simple and user-friendly Task Management System that helps users organize their daily tasks, priorities, and deadlines. The main goal of the project is to make task management more convenient and structured.

MAIN FEATURES

ADD NEW TASKS

SHOW ALL TASKS

FIND TASK

EDIT TASK

MARK TASK AS COMPLETED

DELETE TASKS

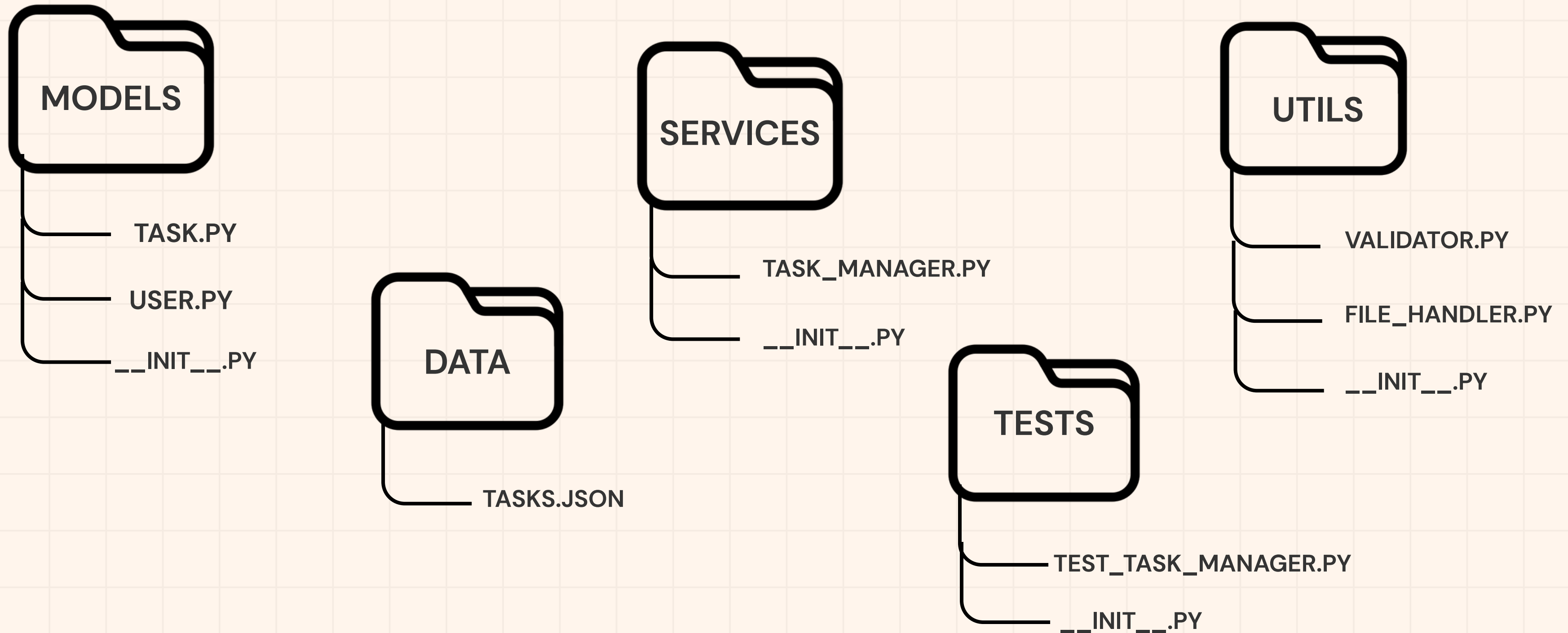
SHOW OVERDUE TASKS

SORT TASKS BY PRIORITY
AND DEADLINE

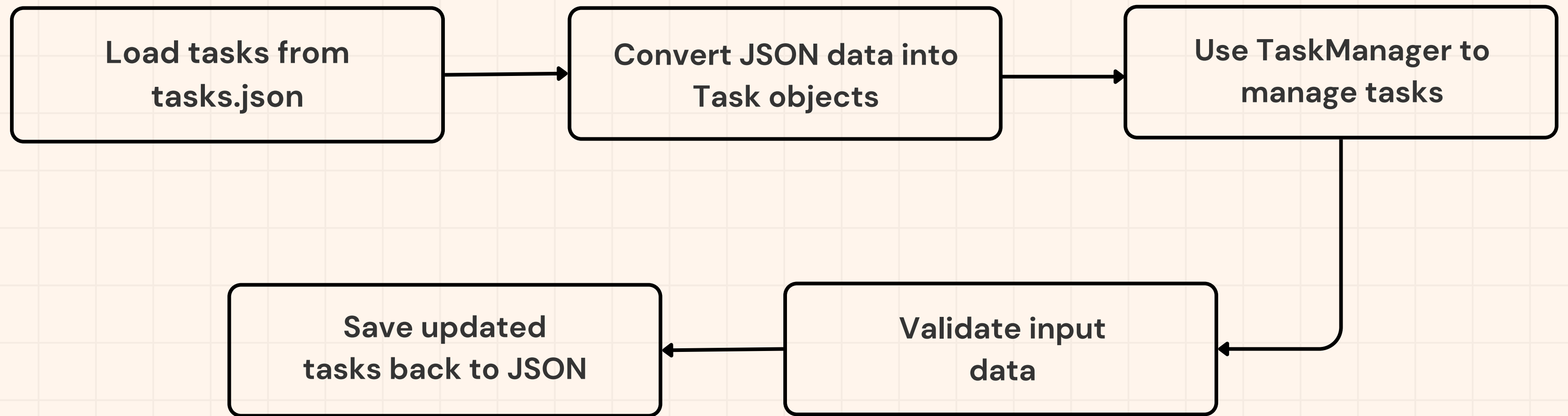
SHOW COMPLETION STATISTICS

HOW DID WE SOLVE IT BRIEFLY?

WE DIVIDED THE PROJECT INTO SEVERAL MODULES TO MAKE THE CODE MORE ORGANIZED AND EASIER TO MAINTAIN.



MAIN LOGIC OF THE SOLUTION



KEY PYTHON CONCEPTS USED

FILE HANDLING

TASKS ARE LOADED FROM JSON
UPDATED TASKS ARE SAVED BACK TO JSON

OOP

TASK CLASS REPRESENTS ONE TASK
USER AND ADMIN CLASSES REPRESENT ROLES
ADMIN INHERITS FROM USER
USER AND ADMIN HAVE DIFFERENT PERMISSIONS

VALIDATION

PRIORITY IS CHECKED
STATUS IS CHECKED
DEADLINE FORMAT IS CHECKED

TESTING

UNIT TESTS CHECK MAIN TASKMANAGER
METHODS

HOW WE ARRANGED THE WORK IN TEAM

DANIYAR

- TASKMANAGER LOGIC
- ADD, EDIT, DELETE, COMPLETE TASKS
- SORTING, OVERDUE TASKS, STATISTICS

AZIZA

- FILE_HANDLER.PY
- JSON LOADING AND SAVING
- TASKS.JSON DATA
- WROTE UNIT TESTS

ALDIYAR

- TASK MODEL
- USER/ADMIN MODEL
- VALIDATORS

MANSUR

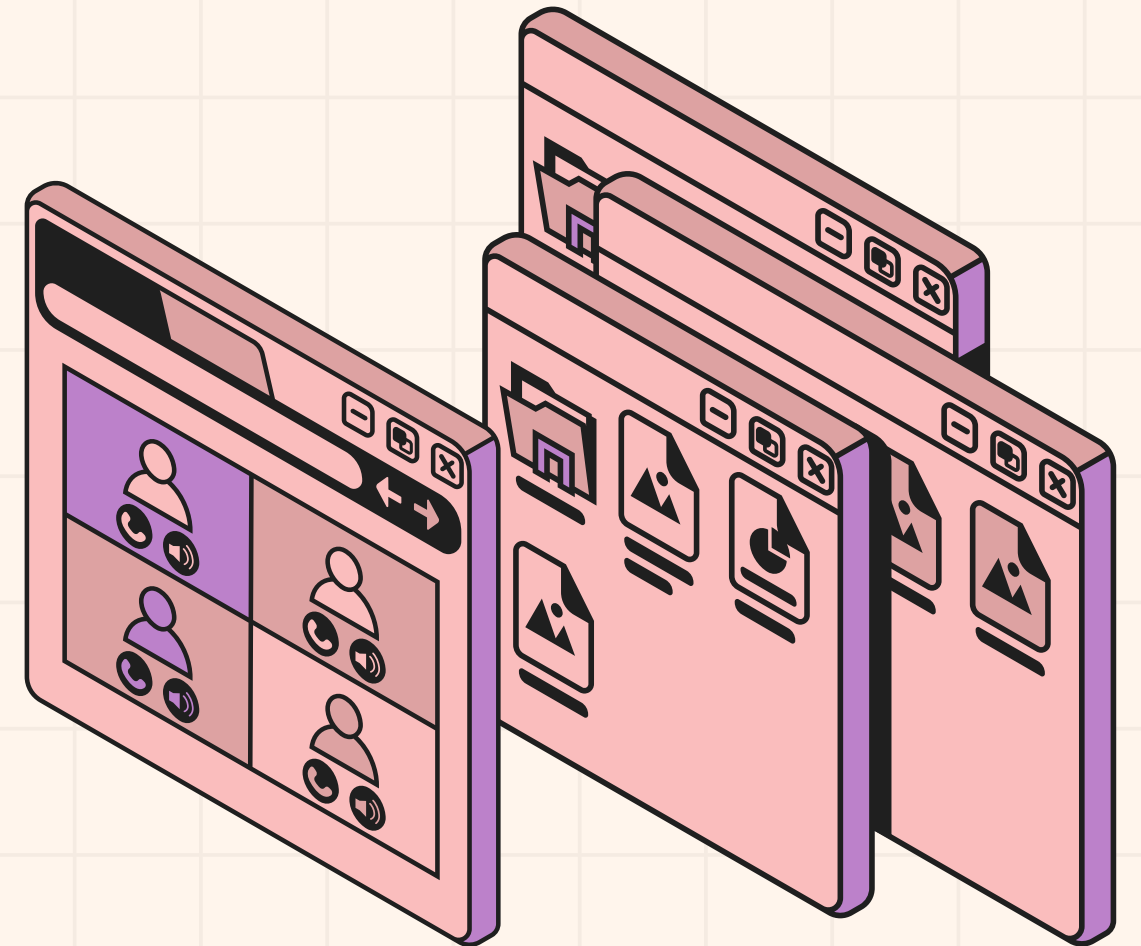
- IMPLEMENTED MAIN.PY
- CREATED THE PROGRAM MENU
- PREPARED README

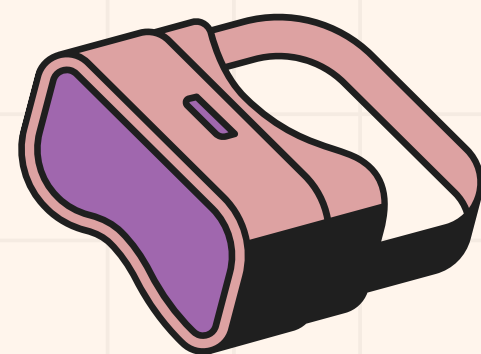
RESULT

```
Welcome to Task Manager!  
1. Login as Admin  
2. Login as User  
Choose role:
```

```
--- MENU ---  
1. View all tasks  
2. View statistics  
3. Add task  
4. Edit task  
5. Delete task  
6. Mark task as completed  
7. Show overdue tasks  
8. Sort tasks by priority  
0. Exit
```

```
--- MENU ---  
1. View all tasks  
2. View statistics  
7. Show overdue tasks  
8. Sort tasks by priority  
0. Exit
```





THANK YOU

